ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–13) | Class 10 Science | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(a)** 12	$3\,Fe + 4\,H_2O \longrightarrow Fe_3O_4 + 4\,H_2$: 3+4+1+4=12.
2	**(d)** increases 100 times	Each unit of pH is a factor of 10.
3	**(d)** silver	Silver lies below hydrogen.
4	**(b)** 13	3n + 1 = 13.
5	**(b)** water	Splitting of water.
6	**(b)** receptor -> sensory neuron -> spinal cord -> motor neuron -> effector	Standard reflex pathway.
7	**(c)** Offspring produced by budding in Hydra	Asexual offspring are clones with minor copying errors.
8	**(b)** 1/2	Offspring Tt and tt in 1 : 1.
9	**(a)** 24 cm	Image forms at C = 2f.
10	**(b)** -1 D	P = 1/(-1 m).
11	**(b)** 2×10^{19}	n = 3.2 / 1.6×10^{-19}.
12	**(d)** is reversed	The field direction reverses.
13	**(c)** small fish	Zooplankton are primary consumers.
14	**(a)** Fe_2O_3	Fe_2O_3 loses oxygen.
15	**(c)** anhydrous sodium carbonate	Loses water of crystallisation.
16	**(d)** They have high melting points	Strong forces between ions.
17	**(b)** $CH_2{=}CH_2$	Dehydration.
18	**(d)** herbivores such as the cow	Cellulose needs long digestion.
19	**(d)** auxin affects stem and root cells in opposite ways	Differential response.
20	**(d)** filament	The anther stalk.
21	**(b)** fertilisation	The sperm type decides.
22	**(b)** 0.5	0.75/1.5 = 0.5.
23	**(d)** a single red patch	Monochromatic light does not disperse.
24	**(b)** 2 ohm	R / n.
25	**(a)** to the south	By Fleming's left-hand rule: thumb points to the south.
26	**(b)** eagle	Top consumer.
27	**(b)** $FeSO_4$ (pale green) is formed	Iron displaces copper.
28	**(c)** neutral	No colour change shows neutral.
29	**(c)** keeps air and water away	Barrier effect.
30	**(d)** forms ionic bonds with chlorine	Carbon forms covalent bonds.
31	**(a)** pulmonary artery	Only artery with deoxygenated blood.
32	**(b)** insulin	Diabetes.
33	**(a)** 23	Gametes have half the number.
34	**(a)** both alleles are recessive	Only tt shows dwarf.
35	**(c)** 40 cm	1/v = 1/20 - 1/40 = 1/40.
36	**(a)** dark	No scattering.
37	**(a)** 1100 W	P = V I.
38	**(b)** increases	Faster change gives a larger emf.
39	**(d)** separating biodegradable from non-biodegradable	For proper treatment.

40	**(b)** 5	$C_3H_8 + 5\,O_2 \longrightarrow 3\,CO_2 + 4\,H_2O$.
41	**(b)** pH 1	Lower pH means more H^+.
42	**(c)** sodium	Only highly reactive metals react with cold water.
43	**(a)** 10	3 x 3 + 1 = 10.
44	**(b)** oxygen	Oxygen is a product.
45	**(a)** relay neuron	Relay neuron.
46	**(a)** Plasmodium	Multiple fission under unfavourable conditions.
47	**(a)** both Tt	3 : 1 ratio arises from Tt x Tt.
48	**(d)** 30 cm	Same size at C.
49	**(a)** +2 D	1/f = 1/25 - 1/50 = 1/50 gives +2 D.
50	**(d)** 2 A	I = 600 / 300.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–13) | Class 10 Science | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(d)** larger than B	Nearer, stronger.
2	**(a)** Plant, caterpillar, bird, hawk	Order of feeding.
3	**(d)** C	Carbon takes oxygen from ZnO.
4	**(c)** thermal decomposition	Heat decomposes baking soda.
5	**(c)** +2	It loses two electrons.
6	**(b)** CH_3COOH	Oxidation.
7	**(a)** cytoplasm	Cytoplasm.
8	**(b)** does not occur	Auxin source is lost.
9	**(d)** unisexual (male)	Only male parts are present.
10	**(a)** 1/2	Each birth is independent.
11	**(a)** 1.5×10^8 m/s	v = c/n.
12	**(b)** the refractive index for violet is higher	n is larger for shorter wavelengths.
13	**(b)** 8 ohm	400 / 50.
14	**(a)** perpendicular to the velocity	Force is always perpendicular to velocity.
15	**(a)** A hawk that eats snakes	Higher level.
16	**(d)** water contains 2 hydrogen atoms for each oxygen atom	$2\,H_2O \longrightarrow 2\,H_2 + O_2$.
17	**(c)** hydrogen	$2\,NaOH + Zn \longrightarrow Na_2ZnO_2 + H_2$.
18	**(c)** is more reactive than iron	Zinc corrodes first.
19	**(c)** high energy is needed to remove 4 electrons	Too much energy.
20	**(b)** 120/80 mm Hg	Standard.
21	**(c)** growth hormone	Pituitary.
22	**(d)** 46 chromosomes	Sperm (23) + egg (23).
23	**(d)** minor DNA copying errors	Copying errors create small differences.
24	**(b)** 3 cm (inverted)	v = 20 cm, m = -1.
25	**(a)** light travels a short path and little blue scatters	Colours combine.
26	**(d)** 484 ohm	V^2 / P.
27	**(b)** the magnet is at rest relative to the coil	No change in the field.
28	**(c)** manure and clean water	Waste management.
29	**(d)** 3	$2\,KClO_3 \longrightarrow 2\,KCl + 3\,O_2$.
30	**(b)** HCl and NaOH	Strong acid + strong base gives neutral salt.
31	**(c)** $Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$	Only a more reactive metal displaces a less reactive.
32	**(a)** 6	4 C-H pairs plus 2 pairs in C=C.
33	**(c)** guard cells take up water and swell	Turgid guard cells open pore.
34	**(b)** muscle	Muscle contracts.
35	**(c)** Inaccurate DNA copying produces variation	Errors in copying provide the basis of variation.
36	**(d)** 40	1/4 of 160.
37	**(a)** 36 cm	1/v = -1/12 + 1/18 = -1/36.
38	**(d)** -40 cm	f = 1/P.
39	**(b)** 36 J	W = V Q.
40	**(a)** a north pole	Anticlockwise looks like north.
41	**(b)** 50 J	Three transfers of 10%.

42	**(d)** thermal decomposition	Heat causes the break-down.
43	**(b)** disappears	Soluble calcium hydrogencarbonate forms.
44	**(d)** X^-	It gains one electron.
45	**(b)** C_2H_4	Unsaturated compounds.
46	**(d)** much more energy per molecule of glucose	Complete breakdown in mitochondria gives more energy.
47	**(d)** abscisic acid	ABA.
48	**(a)** 20 seeds	Each fertilised ovule forms one seed.
49	**(d)** Sperm decides the sex	The father's gamete is X or Y.
50	**(d)** $\sqrt{2}$	sin45/sin30 = (1/$\sqrt{2}$)/(1/2) = $\sqrt{2}$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–13) | Class 10 Science | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(c)** red	n is smallest for red.
2	**(a)** 1 ohm	1/2 + 1/3 + 1/6 = 1.
3	**(d)** zero	No force parallel to the field.
4	**(c)** Destroying the ozone layer	Chlorine destroys ozone.
5	**(a)** prevent oxidation of the oil in the chips	Nitrogen is inert.
6	**(a)** NaOH	NaOH ionises completely in water.
7	**(d)** anodising	Anodising.
8	**(b)** 6 protons cannot hold 10 electrons easily	Nucleus can't hold extra electrons.
9	**(c)** xylem	Xylem.
10	**(c)** iodine	Goitre.
11	**(d)** umbilical cord	It carries nutrients to the foetus.
12	**(d)** an acquired trait due to environment	Environment affects the phenotype.
13	**(c)** 20 cm	1/u = 1/v - 1/f = 1/60 - 1/15 = -1/20.
14	**(d)** blue light	Short wavelength scatters more.
15	**(a)** 1 kWh	2 x 0.5.
16	**(b)** a larger induced emf	More turns, more emf.
17	**(d)** converting vegetable waste into manure	Natural decomposition.
18	**(c)** 9	$4\,Al + 3\,O_2 \longrightarrow 2\,Al_2O_3$: 4+3+2=9.
19	**(c)** basic	Strong base and weak acid give basic salt.
20	**(a)** Mg > Zn > Fe > Cu	Standard reactivity series.
21	**(c)** 5	2 C-H plus 3 in triple bond.
22	**(a)** no blue-black colour	No starch is made in dark.
23	**(d)** dendrite to axon end	Direction.
24	**(a)** vegetative propagation	Buds in the eyes sprout.
25	**(c)** TT	A homozygous genotype forms a single type of gamete.
26	**(b)** 12 cm behind the mirror	1/v = 1/20 + 1/30 = 1/12.
27	**(b)** in front of the retina	Because the eye is too strong.
28	**(a)** 10 V	V = I R.
29	**(d)** increases	Proportional to the current.
30	**(b)** 10,000 J	10 x 1000.
31	**(a)** $CaO + H_2O \longrightarrow Ca(OH)_2$	Quick lime slaking releases heat.
32	**(b)** formation of gypsum	Water is taken up.
33	**(c)** $MgCl_2$	Mg^{2+} and Cl^-.
34	**(a)** addition reaction	Double bond opens.
35	**(a)** lactic acid accumulation	Anaerobic respiration.
36	**(b)** cytokinin	Cell division.
37	**(c)** self-pollination	Same plant, so genetically it counts as self-pollination.
38	**(b)** 23	22 + 1 = 23.
39	**(a)** water	Water has a lower n.
40	**(c)** white light	Recombination.
41	**(d)** 1 A	I = 16 / 16.
42	**(a)** Reversing the current	Direction changes.

43	**(d)** UV rays of the sun	Protects from skin cancer.
44	**(d)** is more reactive than iron and forms a protective layer	Zinc protects the iron.
45	**(b)** sulphuric acid	Lead-acid batteries.
46	**(d)** heating in air	Heating in air promotes oxidation.
47	**(a)** no free electrons	All valence electrons used in bonds.
48	**(b)** transpiration pull	Pull on water column.
49	**(d)** adrenaline	Adrenal gland.
50	**(a)** In the urethra / vas deferens region	Semen is formed before ejaculation.

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